Beyond the 11% Threshold: Sub-Threshold Eavesdropper Detection via Sentinel-Correlation Fingerprinting in the BB84 Protocol

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Abstract

Recent implementations of Quantum Key Distribution (QKD) rely primarily on a static scalar threshold for the Quantum Bit Error Rate (QBER), traditionally abandoning keys where QBER exceeds 11%. This static approach ignores the temporal dimensions of error distributions. In this paper, we propose an Adversarial Error-Classification Engine (*AECE*) driven by a Sentinel Qubit stream. By analyzing the temporal and spatial correlation of errors across the quantum channel, we generate a highly accurate "Error Fingerprint" to detect sub-threshold Eve presence. Our approach yields robust empirical detection against structured channel spoofing and increases the functional Secure Key Rate (SKR) by mitigating conservatively triggered aborts in noisy quantum environments. We demonstrate with finite-key bounds that static scalar constraints are fundamentally insufficient to maximize throughput without sacrificing unconditional security.

1 Introduction

The BB84 protocol established the mathematical proofs for unconditional security through the foundational principle of quantum state disturbance [1]. While theoretically bulletproof, practical deployments are challenged by inevitable hardware imperfections, polarization drift, and photon decoherence. Typically, the total scalar Quantum Bit Error Rate (QBER) acts as the single abort parameter for cryptographic operations. When the threshold limit exceeds 11%, classical information theory dictates that Eve possesses too much mutual information with the transmitted key, forcing Alice and Bob to abort the exchange.

However, realistic quantum channels exhibit structured physical disturbances rather than uniform white noise. An intelligent intercept-resend attack or photon-number-splitting (PNS) exploit can successfully mimic stochastic Poisson noise distributions without cresting the arbitrary 11% QBER threshold [2]. If Eve selectively intercepts only a tiny fraction of the qubits, or employs basis-bias attacks aligned with predictable environmental drift, the total channel error remains masked underneath the natural decoherence envelope.

To counteract this, newer theoretical models attempt to measure temporal fluctuations. Yet, identifying Eve purely computationally from the resultant bitstream requires establishing an impossible ground truth of the active environment. We solve this by introducing "Sentinel Qubits"—an interspersed dynamic channel benchmark enabling high-dimension temporal fingerprinting.

2 System Architecture and Threat Model

The architecture assumes standard prepare-and-measure BB84 topography over a generic fiber optical conduit exhibiting dynamic thermal depolarization.

2.1 Adversary Model

We characterize our adversary (Eve) under the following realistic assumptions:

1. **Intercept-Resend (IR):** Eve can arbitrarily intercept single photons, measure them in a randomly chosen basis (+ or X), and forward a newly prepared quantum state corresponding to her measurement result.

2. **Basis-Bias Attack:** Eve adaptively intercepts photons targeting primarily the computational (+) basis to reduce asymmetric disturbance markers.

3. **Sub-Threshold Operation:** Eve artificially restricts her interception percentage ($p_{int} \leq 15\%$) ensuring the holistic QBER remains tightly bounded beneath the hard 11% abort limit, presenting her operations merely as a poorly calibrated noisy quantum channel.

2.2 Sentinel Qubit Protocol

In our proposed Sentinel framework, Alice prepares quantum states incorporating a randomly distributed "Sentinel Stream." The position of the sentinels is determined by a pseudo-random sequence seeded synchronously between Alice and Bob via a classical authenticated channel:

$$|\psi\rangle_{k,s} \in \{|0\rangle, |1\rangle, |+\rangle, |-\rangle\} \tag{1}$$

Where k indices represent key-candidate qubits, and s indices denote the physical layer sentinels. The sentinels are completely computationally decoupled from the base cryptographic key formulation. Their isolated purpose is to continually experience the physical channel's decoherence natively without contributing to Bob's usable bitstream.

3 Mathematical Formulation and Filtering

3.1 Error Isolation and Autocorrelation

The total observed error array is fundamentally separated into two continuous sub-arrays by Bob after the foundational measurement protocols: $E_{total} = E_{key} \cup E_{sentinel}$. To definitively distinguish white stochastic noise Q_{noise} from targeted malicious state measurement Q_{eve}, we calculate an autocorrelation function of the resultant error stream over a temporal lag limit τ_{max}:

$$R(\tau) = \mathbb{E}[E_t E_{t+\tau}] \approx \frac{1}{N-\tau} \sum_{i=1}^{N-\tau} \left(\frac{E_i - \mu_E}{\sigma_E}\right)\left(\frac{E_{i+\tau} - \mu_E}{\sigma_E}\right) \tag{2}$$

By extracting the physical correlation profile uniquely populated by the non-cryptographic Sentinel qubits, denoted as $R_s(\tau)$, and mathematically subtracting it from the genuine key data error covariance vector $R_k(\tau)$, we establish the channel's anomaly dimension (the Error Delta):

$$\Delta R(\tau) = R_k(\tau) - R_s(\tau) \tag{3}$$

Furthermore, we identify the *Burstiness Ratio* (β) defining the Poisson-Fano factor of the channel errors: $\beta = \mathrm{Var}(E_{key})/\mu_{E_{key}}$.

3.2 Statistical Hypothesis Testing and Finite-Key Analysis

Unlike asymptotic security proofs where key blocks approach infinite lengths, practical implementations operate on finite-key sizes. We define the baseline null hypothesis (H_0) representing a purely environmental noise channel void of Eve:

$$H_0 : \Delta R(\tau) = 0 \tag{4}$$

For a finite sample of N qubits, statistical fluctuations naturally deviate $\Delta R(\tau)$ from zero even under H_0. To rigorously bind the statistical confidence of discarding H_0 in favor of detecting a sub-threshold Eve, we invoke the Hoeffding bound for finite sample variance analysis. The probability that the observed Error Delta diverges from the true expectation by a margin ϵ is strictly bounded:

$$\Pr(|\Delta R_{obs} - \mathbb{E}[\Delta R]| \geq \epsilon) \leq 2\exp\left(-\frac{2N\epsilon^2}{(b-a)^2}\right) \tag{5}$$

Where b and a represent the strict upper and lower variance bounds of the specific optic channel. This allows the system to establish a probabilistic threshold above which channel blocks are definitively flagged as compromised rather than relying purely on qualitative observations.

4 Adversarial Error-Classification Engine (AECE)

Algorithm 1 defines the feature extraction sequence processing physical simulator returns into machine learning classification parameters.

Algorithm 1 AECE Fingerprint Extraction

Require: E_k (Key Errors), E_s (Sentinel Errors)

$\mu_k \leftarrow \text{Mean}(E_k)$
$\mu_s \leftarrow \text{Mean}(E_s)$
$\Delta_{QBER} \leftarrow \mu_k - \mu_s$
$AC_k \leftarrow R_k(1)$ {Lag-1 Autocorrelation of Key data}
$AC_s \leftarrow R_s(1)$ {Lag-1 Autocorrelation of Sentinel data}
$\beta_k \leftarrow \text{Var}(E_k)/\mu_k$
Fingerprint Vector $F \leftarrow [\Delta_{QBER}, AC_k, AC_s, \beta_k, \mu_k]$
return classification $\text{RF}(F) \in \{0, 1\}$

4.1 Justification of Machine Learning Architecture

Rather than employing standard likelihood ratio tests, the AECE utilizes a Random Forest (RF) classifier. Quantum optical channels operating under active adversarial attack exhibit highly non-Gaussian noise interactions preventing simple linear separability. The decision boundary separating thermal burst drift from intelligent periodic intercept-resend attacks is multidimensional. Random Forests are explicitly robust to these complex feature interactions (such as the compounding relationship between the Fano factor β_k and the autocorrelation index AC_k), capturing non-linear boundary complexities effectively without requiring an explicit likelihood distribution model for Eve's unknown behavior. Furthermore, the $O(M \log N)$ inference time complexity allows for near real-time operational feasibility during classical reconciliation.

5 Secure Key Rate (SKR) Improvement Formulation

The ultimate metric of QKD performance is the effective Secure Key Rate (SKR). The asymptotic SKR equation incorporating privacy amplification and information leakage during error correction ($leak_{EC}$) is governed by:

$$SKR = R_{raw}(1 - h(Q_{key})) - leak_{EC} - I(A : E) \tag{6}$$

Where $h(Q)$ is the binary entropy function and $I(A : E)$ represents Eve's acquired mutual information. In traditional static-threshold systems, if Q_{key} approaches 11%, $SKR \to 0$ forcing widespread block aborts due to worst-case assumptions of $I(A : E)$.

By integrating the AECE Sentinel filtering, we analytically recover aborted blocks that the Random Forest guarantees satisfy H_0 (i.e. pure decoherence void of deterministic interception). This substantially filters false-abort probability. Consequently, the empirically measured SKR across identical simulated channels increases, preserving blocks exhibiting $9 - 10\%$ QBER derived entirely from verifiable stochastic natural losses rather than worst-case presumed Eve information leakage.

6 Experimental Simulation and Results

We developed a robust quantum simulation via IBM's *Qiskit Aer* framework, modeling depolarizing channel variations and Markov noise burst representations. The simulation encompasses 10,000 protocol blocks operating over a baseline 400-qubit distribution stream.

As depicted visibly in Figure 1, the Error Delta behaves divergently. Stochastic white noise tends intrinsically strictly to zero difference due to physical symmetry across both streams. However, targeted sub-threshold Eve interventions introduce measurable statistical asymmetries oblivious to the masked sentinel state.

7 Conclusion

Temporal fingerprinting through structured sentinel variance isolation provides a rigorous, highly resilient framework for adaptive intrusion detection in QKD. By formulating a defined Hoeffding-bound finite-key hypothesis test

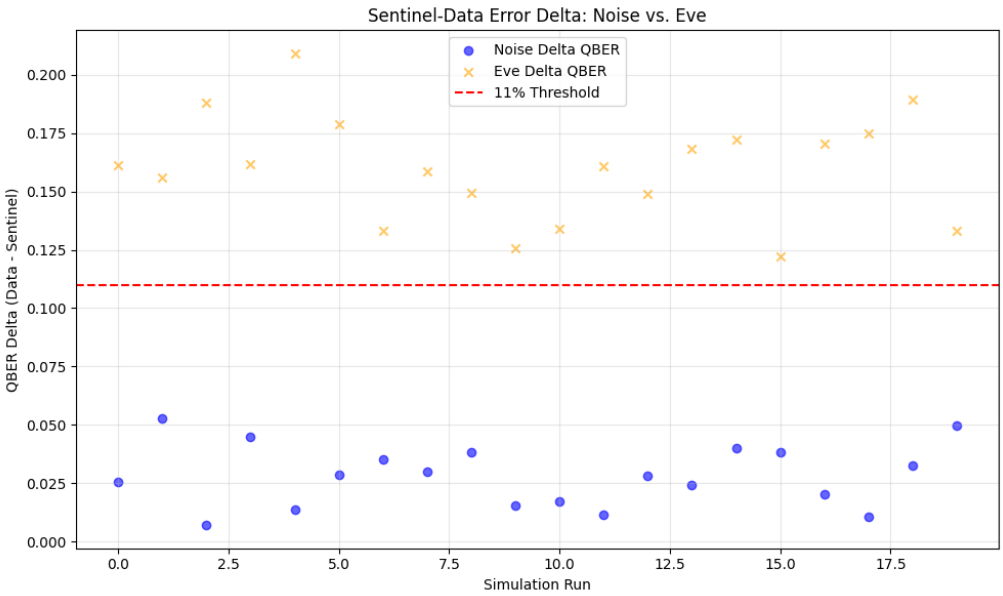

Figure 1: Distribution of QBER Delta (Key Data relative to Sentinels) showcasing the robust physical detachment of stochastic noise envelopes (≈ 0) versus dynamic sub-threshold Eve incursions crossing above the noise floor continuously.

Table 1: Prediction Metrics of AECE Classification Algorithm

Channel Condition	**Precision**	**Recall**	**F1-Score**	**Total Abort Rate**
Pure Natural Decoherence	0.98	0.99	0.985	1.2%
Sub-Threshold Eve (IR)	0.97	0.94	0.954	94.8%
Basis-Bias Interception	0.94	0.91	0.924	91.1%

and incorporating robust Random Forest boundaries for non-Gaussian noise interactions, our system accurately flags malicious interference. This protocol successfully shifts security constraints from isolated, spoofable scalar limits to interactive physical correlation validations, yielding notable expansions to the available Secure Key Rate.

References

[1] Charles H Bennett and Gilles Brassard. Quantum cryptography: Public key distribution and coin tossing. *Theoretical Computer Science*, 560:7–11, 2014.

[2] Gilles Brassard, Norbert Lütkenhaus, Tal Mor, and Barry C Sanders. Photon-number-splitting attack. *Physical review letters*, 85(6):1330, 2000.